FRAP Analysis Postprocessing Pipeline

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1 Overview

The FRAP post-analysis pipeline transforms raw target and control fluorescence measurements into corrected recovery curves, summarizes them at the group level, fits a one-phase exponential recovery model, and derives quantitative FRAP parameters such as recovery half-time, mobile fraction, and immobile fraction.

The final postprocessing approach consists of a single normalization strategy. First, the target signal is corrected for acquisition-related photobleaching using the control signal. Second, the corrected target signal is projected onto a recovery percentage scale where the corrected pre-bleach intensity is set to 100% and the corrected immediate post-bleach intensity is set to 0%.

The complete processing chain can be summarized as

$$\text{Raw target and control intensities} \rightarrow$$
$$\rightarrow \text{Control-based acquisition correction} \rightarrow \text{Pre/post recovery normalization} \rightarrow$$
$$\rightarrow \text{Group recovery summary} \rightarrow \text{Exponential recovery fit} \rightarrow$$
$$\text{FRAP parameters.}$$

2 Input Intensities

For each sample, the input dataframe contains a time series of target and control fluorescence intensities. The target intensity is denoted by

$$T(t),$$

and the control intensity is denoted by

$$C(t).$$

The target trace $T(t)$ corresponds to the bleached FRAP region of interest. The control trace $C(t)$ corresponds to an unbleached reference region and is used to estimate acquisition-related intensity loss.

The pre-bleach frame is denoted by

$$t_0,$$

and corresponds in the implementation to

$$t_0 = \texttt{PREBLEACH_TIME}.$$

The immediate post-bleach frame is denoted by

$$t_1,$$

and is defined as the first frame satisfying

$$t_1 > t_0.$$

If no pre-bleach frame or no immediate post-bleach frame exists, the sample cannot be processed and is rejected.

3 Control-Based Acquisition Correction

The control trace is used to estimate global acquisition-related photobleaching or intensity drift during imaging. First, the control intensity at the pre-bleach frame is extracted:

$$C_0 = C(t_0).$$

The acquisition correction factor is then computed at each frame:

$$B(t) = \frac{C(t)}{C_0}.$$

This factor describes how much of the control signal remains relative to the pre-bleach control intensity. For example, $B(t) = 0.9$ indicates that the control signal has decreased to 90% of its pre-bleach value.

The target signal is corrected by dividing the raw target intensity by this factor:

$$T_{\text{corr}}(t) = \frac{T(t)}{B(t)}.$$

Equivalently,

$$T_{\text{corr}}(t) = \frac{T(t)}{C(t)/C_0} = T(t)\frac{C_0}{C(t)}.$$

This produces a target trace corrected for acquisition-related intensity loss estimated from the control region. If $C_0 = 0$, or if C_0 is not finite, the correction is undefined and the sample is rejected.

4 Pre/Post Recovery Normalization

After acquisition correction, the corrected pre-bleach target intensity is defined as

$$I_{\text{pre}} = T_{\text{corr}}(t_0),$$

and the corrected immediate post-bleach target intensity is defined as

$$I_{\text{post}} = T_{\text{corr}}(t_1).$$

The corrected target trace is then projected onto a pre/post recovery percentage scale:

$$R(t) = \frac{T_{\text{corr}}(t) - I_{\text{post}}}{I_{\text{pre}} - I_{\text{post}}} \cdot 100.$$

Therefore,

$$R(t_0) = 100,$$

and

$$R(t_1) = 0.$$

The resulting quantity $R(t)$ is the corrected recovery percentage. It represents recovery relative to the amount of signal lost during bleaching. A value of 0% corresponds to the immediate post-bleach corrected target intensity, and a value of 100% corresponds to the corrected pre-bleach target intensity.

If

$$I_{\text{pre}} - I_{\text{post}} \leq 0,$$

or if the denominator is not finite, the recovery percentage is undefined for that sample.

5 Time Filtering

Only frames from the configured minimum plotting time onward are retained:

$$t \geq t_{\min},$$

where

$$t_{\min} = \texttt{MIN_PLOT_TIME}.$$

This step removes frames that should not be included in plotting or downstream recovery analysis. The x-axis in the plots represents frame index unless the input time column has been explicitly converted into physical time units.

6 Group Recovery Summary

For each experimental group, the normalized recovery signal $R(t)$ is aggregated across samples. At each frame, the group mean recovery curve is computed as

$$\bar{R}(t) = \frac{1}{N_t} \sum_{j=1}^{N_t} R_j(t),$$

where N_t is the number of samples contributing to that frame.

The percentile outline is computed from the same per-sample recovery traces. For each frame, the values $R_j(t)$ across samples are collected and the chosen lower and upper percentiles are computed. For example, the 25th and 75th percentile curves are

$$P_{25}(t) = Q_{0.25}\left(\{R_j(t)\}_{j=1}^{N_t}\right),$$

$$P_{75}(t) = Q_{0.75}\left(\{R_j(t)\}_{j=1}^{N_t}\right).$$

The group mean curve $\bar{R}(t)$ is the curve fitted by the exponential recovery model. Individual traces do not need to be shown in the final plot, but they are still used to compute the group mean and percentile outline.

7 One-Phase Exponential Recovery Model

The group mean FRAP recovery curve is modeled using a one-phase exponential recovery function:

$$I(t) = I_\infty - Ae^{-kt}.$$

Here, I_∞ is the fitted recovery plateau, A is the fitted recovery amplitude, and k is the fitted recovery rate constant.

The fitting time axis is shifted so that the first fitted frame corresponds to zero:

$$t_{\text{start}} = \min(x_i),$$

$$t_i = x_i - t_{\text{start}},$$

where x_i are the original frame values.

The fitted model is therefore

$$\bar{R}_i \approx I_\infty - Ae^{-kt_i}.$$

Because the fitted signal is a recovery percentage, I_∞, A, and fitted intensity values are expressed in percentage units.

8 Initial Parameter Estimates

Before nonlinear curve fitting, the initial parameter vector is estimated from the group mean recovery curve. The initial plateau estimate is the final observed group mean recovery value:

$$I_\infty^{(0)} = \bar{R}_{\text{last}}.$$

The initial amplitude estimate is

$$A^{(0)} = \max\left(I_\infty^{(0)} - \bar{R}_0, 10^{-6}\right).$$

The initial recovery rate constant is set to the configured value:

$$k^{(0)} = k_{\text{init}}.$$

Thus, the initial parameter vector is

$$\boldsymbol{\theta}^{(0)} = \begin{bmatrix} I_\infty^{(0)} \\ A^{(0)} \\ k^{(0)} \end{bmatrix}.$$

9 Parameter Bounds

The nonlinear fit constrains the parameters to physically meaningful ranges:

$$I_\infty \geq 0,$$

$$A \geq 0,$$

$$k \geq k_{\min}.$$

10 Fitted Initial Post-Bleach Recovery

At shifted time $t = 0$, the model gives

$$I(0) = I_\infty - A.$$

The fitted initial post-bleach recovery value is therefore

$$I_0 = I_\infty - A.$$

Because the per-sample traces are normalized so that the measured immediate post-bleach frame is 0%, I_0 is expected to be close to 0% when the immediate post-bleach frame is included in the fitted data. It may differ from zero because the fit is performed on the group mean curve and is constrained by all fitted frames.

11 Recovery Half-Time

The recovery half-time is the time required for the fitted recovery curve to recover half of its amplitude. For

$$I(t) = I_\infty - Ae^{-kt},$$

the half-time is

$$t_{1/2} = \frac{\ln 2}{k}.$$

A larger value of k corresponds to faster recovery and therefore to a smaller value of $t_{1/2}$.

The half-time on the original, unshifted frame axis is

$$t_{1/2}^{\text{abs}} = t_{\text{start}} + t_{1/2}.$$

12 Half-Recovery Intensity

The intensity at half recovery is halfway between I_0 and I_∞:

$$I_{1/2} = I_0 + \frac{A}{2}.$$

Since

$$A = I_\infty - I_0,$$

this can also be written as

$$I_{1/2} = \frac{I_0 + I_\infty}{2}.$$

13 Mobile Fraction

In the final normalized recovery scale, the pre-bleach reference intensity used for mobile fraction estimation is fixed to

$$I_{\text{pre,fit}} = 100.$$

The total bleached amount in the fitted recovery scale is therefore

$$I_{\text{pre,fit}} - I_0 = 100 - I_0.$$

The recovered amount is the fitted recovery amplitude:

$$A = I_\infty - I_0.$$

The mobile fraction is therefore

$$M_f = \frac{A}{100 - I_0}.$$

Equivalently,

$$M_f = \frac{I_\infty - I_0}{100 - I_0}.$$

The mobile fraction is clipped to the interval

$$0 \le M_f \le 1.$$

If

$$100 - I_0 \le 0,$$

or if the denominator is not finite, then the mobile and immobile fractions are undefined.

When I_0 is close to 0%, the mobile fraction is approximately

$$M_f \approx \frac{I_\infty}{100}.$$

14 Immobile Fraction

The immobile fraction is defined as the unrecovered fraction:

$$I_f = 1 - M_f.$$

A high mobile fraction indicates that most of the bleached signal recovered, whereas a high immobile fraction indicates incomplete recovery.

15 Smooth Fitted Recovery Curve

For visualization, the fitted model is evaluated over a dense set of shifted frame values:

$$t_{\text{smooth}} \in [\min(t_i), \max(t_i)].$$

The fitted recovery values are

$$I_{\text{fit}}(t_{\text{smooth}}) = I_\infty - Ae^{-kt_{\text{smooth}}}.$$

The smooth frame axis is then shifted back to the original frame coordinates:

$$x_{\text{smooth}} = t_{\text{smooth}} + t_{\text{start}}.$$

The output fitted curve is therefore represented as

$$\left(x_{\text{smooth}}, I_{\text{fit}}(t_{\text{smooth}})\right).$$

16 Final Output Per Group

For each group, the pipeline returns the following quantities:

Quantity	Meaning
group	Group identifier
$I_{\text{pre,fit}}$	Pre-bleach reference intensity; fixed to 100
I_0	Fitted initial post-bleach recovery value
I_∞	Fitted recovery plateau in percent
A	Fitted recovery amplitude in percent
k	Fitted recovery rate constant
$t_{1/2}$	Recovery half-time on the shifted fitting frame axis
$t_{1/2}^{\text{abs}}$	Recovery half-time on the original frame axis
$I_{1/2}$	Recovery value at half recovery
M_f	Mobile fraction
I_f	Immobile fraction
n_{files}	Number of unique source files contributing to the group

Table 1: Summary of output quantities generated for each fitted group.

17 Conceptual Interpretation

The analysis estimates how fluorescence in a bleached target region recovers over time after correcting for global acquisition-related intensity loss using a control region.

The control signal is used only for correction:

$$C(t) \to B(t) = \frac{C(t)}{C_0}.$$

The target signal is corrected using this factor:

$$T(t) \to T_{\text{corr}}(t) = \frac{T(t)}{B(t)}.$$

The corrected target signal is then normalized using its own corrected pre-bleach and immediate post-bleach values:

$$T_{\mathrm{corr}}(t) \to R(t) = \frac{T_{\mathrm{corr}}(t) - I_{\mathrm{post}}}{I_{\mathrm{pre}} - I_{\mathrm{post}}} \cdot 100.$$

The corrected recovery curves are summarized within each group:

$$\{R_j(t)\} \to \bar{R}(t), P_{25}(t), P_{75}(t).$$

The group mean curve is fitted using a one-phase exponential recovery model:

$$\bar{R}(t) \to I(t) = I_\infty - Ae^{-kt}.$$

Finally, the fitted model is used to estimate recovery kinetics and recovery extent:

$$\text{Exponential fit} \to \left\{I_0, I_\infty, A, k, t_{1/2}, M_f, I_f\right\}.$$